

Sam Alexander

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Summary:

API-to-dashboard data engineer with extensive experience in owning and managing data and data infrastructure. Successfully transitioned ETL/ELT orchestration from AWS Lambda to Airflow, while completely redesigning the ingestion methodology to increase clarity and make everything easier to maintain.

Technical Skills:

- **Programming Languages:** Python (Advanced), SQL (Advanced), R (Advanced), Bash (Intermediate)
- **Data Engineering Tools:** Airflow (Advanced), Snowflake (Advanced), Redshift (Advanced), Power BI (Advanced), Grafana (Intermediate), Spark (Basic via Personal Projects), Databricks (Basic via Personal Projects), Kubernetes (Basic)
- **Cloud Platforms:** AWS (Advanced, S3, EC2, RDS, Lambda), Terraform (Intermediate)
- **Other Tools:** Git (Intermediate), BitBucket (Intermediate), Excel (Advanced, Data Analysis)

Work Experience:

Data Engineer. *Qortex*, May 2024 – Present

- **Primary accomplishments:**
 - Maintained and optimized Airflow installation, reducing overall DAG run times and increasing data consistency.
 - Migrated infrastructure monitoring from Datadog to Grafana to significantly reduce monthly costs
 - Consolidated Grafana reporting dashboards to enable users to switch between reports with dropdowns instead of having to navigate away and back again.
 - Reduced Snowflake storage costs by at least 25% by modifying table configurations and removing redundant columns
- **Programming languages used:** Python, SQL, R for local exploration
- **Technology used:** Snowflake, AWS (S3, EC2), Grafana, Airflow, Terraform, DynamoDB, Docker (Colima), MacOS

Data Engineer. *Enthusiast Gaming*, Aug 2019 – May 2024

- **Primary accomplishments:**
 - Migrated ingestion pipeline from AWS Lambda to Airflow to increase observability and significantly reduce cognitive complexity
 - Redesigned ingestion pipeline, switching from ETL to ELT. This both simplified the

- data model and fixed the raw data backup in S3.
- Mentored the 1-2 data engineers under me while also being responsible for creating Jira tickets and assigning work to the team each week
- **Programming languages used:** Python, SQL, R for local exploration
- **Technology used:** Redshift, MySQL, PostgreSQL, AWS (S3, EC2, RDS, IAM, Secrets Manager, Athena, Glue), Power BI, Airflow, Google Ad Manager, Kubernetes, Linux, MacOS, Bitbucket

Database Developer. *Vretta Inc.*, Nov 2017 – Apr 2019

- **Programming languages used:** SQL, R, Node.js
- **Technology used:** MySQL, PostgreSQL, AWS (EC2, RDS, Route 53), Docker

Data Analyst. *Orbit Group Partners*, Jun 2014 – Nov 2017

- **Programming languages used:** SQL, R, VBA
- **Technology used:** Microsoft SQL Server, Excel, Powerpoint

Education:

MA, Mathematics, York University, 2013–14

BSc, Mathematics (Honours), Trent University, 2009–13